Sukhrob Ilyosbekov

Machine Learning Engineer · Computer Vision & LLM Systems

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Summary

I'm a machine learning engineer who wrote production software for years before moving into ML, so I care just as much about what happens after training, when the model is serving real users and the data starts to drift. Most of my current work is on healthcare data. My research is in computer vision, mainly medical imaging and image-text models for biology. I'm comfortable with the math behind the methods and enjoy turning a paper into code that actually runs.

Technical Skills

Deep Learning & CV: PyTorch, TensorFlow, CNNs, Vision Transformers, DINOv3, EfficientNet, YOLO, contrastive learning, GradCAM++, OpenCV

LLM Systems: Claude/GPT/DeepSeek APIs, tool-use agents, MCP, RAG, vision-language models, Hugging Face

Math & Modeling: Probability and statistics, linear algebra, optimization, time-series forecasting, feature engineering, A/B testing, error analysis

Languages: Python, C#, C++, TypeScript, Kotlin

Data & Serving: NumPy, pandas, scikit-learn, FastAPI, PostgreSQL, SQL Server, MongoDB, Redis, OHIF/DICOM

Infrastructure: CUDA, mixed-precision and multi-GPU training, model serving and drift monitoring, Docker, Kubernetes, GitHub Actions, AWS, Azure

Software: ASP.NET Core, SignalR, Blazor, Node.js, Bun, NestJS, React, Next.js, Angular

Education

Northeastern University, M.S. in Computer Science, GPA 4.0, Boston, MA — *Jan 2024 – May 2026*

Suffolk University, B.S. in Computer Science, Cum Laude, Boston, MA — *Sep 2021 – May 2023*

Honors: National Physics Olympiad, 1st Place (Uzbekistan) · Associate Member, Sigma Xi · Coding Competition Winner, TUIT

Professional Experience

Generative AI Engineer | Sidebench | Los Angeles, CA — *Jun 2026 – Present*

- Build the evaluation harnesses that decide whether off-the-shelf image and language models can do a client's job, with resumable runs, per-call costs, and reports that show every result.
- Cosmetic surgery previews: runs facelift, rhinoplasty, eyelid and BOTOX edits through about ten hosted and self-hosted image models. Landmark masks confine each edit to the treated area; ArcFace checks identity.
- Social stories for autistic children: compared four image models and five LLMs (Claude, GPT, DeepSeek) to illustrate and write the stories, scored by automated checks on a draft content standard and a human rubric.

Machine Learning Engineer | EmTech Care Labs | Portland, ME — *Jan 2025 – Jun 2025*

- Built a model that reads Alzheimer's residents' daily activity and caregiver logs and flags who is at risk of decline, so staff can step in before a crisis.
- Wrote an NLP pipeline that turns free-text clinical notes into structured records of symptoms, medications, and care events. It replaced manual chart review.
- Both models run in production on de-identified data under HIPAA, with nightly retraining, drift alerts, encrypted PHI, and an audit trail on every prediction.
- Also led engineering for the care platform itself, a TypeScript monorepo on AWS that passed an outside HIPAA readiness review with no critical findings.

Machine Learning Engineer | AllFactors | Santa Clara, CA *Oct 2021 – Dec 2024*

- Built the property-valuation and market-forecast models that replaced the company's rules-based system.
- Wrote the real-time feature pipeline behind them on SignalR and SQL Server change tracking. New transactions reached the models within seconds and fed a nightly retrain.
- Users saw predictions in Blazor WebAssembly dashboards; automated tests covered the whole path from incoming data to prediction.

Earlier: Software Engineer at Smart Meal Service, Russia (2020 – 2021), writing C# software for food kiosks and a robotic cashier. Game Developer at Pentalight Technology, Malaysia (2020 – 2021), on networking and hand tracking for a Unity VR simulation. Freelance Python NLP and .NET work (2019 – 2020). Game Developer at EC Dev Team, Uzbekistan (2016 – 2019).

Selected AI/ML Projects

MorphoCLIP GitHub | Paper
PyTorch, DINOv3, BioClinical ModernBERT, contrastive learning, CPJUMP1

- Matches Cell Painting microscopy of cells treated with drugs or genetic perturbations to plain-language descriptions of the treatment, using contrastive image-text training.
- The DINOv3 and BioClinical ModernBERT backbones stay frozen and only the projection heads train, so it fits on one consumer GPU.
- Evaluated on CPJUMP1 (51 plates, 3M+ images). As first author I led design, training, and evaluation.

Localize, Don't Beautify https://arxiv.org/abs/2608.02841
Image-editing APIs, inpainting, ArcFace, prompt engineering

- Commercial image editors asked for one surgical change tend to retouch the whole face. I tested prompt steering, masked compositing, and inpainting as client-side fixes across six editors on 196 preview edits.
- Scored identity preservation with ArcFace and checked whether each edit stayed in its region. Plain masked compositing kept edits in place best. Sole author.

MelanomaNet GitHub | Paper
PyTorch, EfficientNet V2, focal loss, GradCAM++, OpenCV, ISIC 2019

- Nine-class skin-lesion classifier on ISIC 2019 (25K images). EfficientNet V2 with focal loss to handle the class imbalance; 0.86 weighted F1.
- Scores the model's GradCAM++ attention against the ABCDE criteria dermatologists use, each measured from the lesion mask. Sole author.

Med Image Scanner https://github.com/suxrobgm/med-image-scanner
FastAPI, PyTorch, OpenCV, OHIF Viewer, DICOM, Next.js

- Pulls X-ray, CT, and MRI studies from hospital imaging systems over DICOM and runs PyTorch detectors on them, such as pneumonia on chest X-rays and hemorrhage on head CTs.
- Results show up as overlays in OHIF. Studies are de-identified on the fly, and access is role-based and logged.

LogisticsX https://logisticsx.app | GitHub
Multi-provider LLMs, MCP, .NET 10, EF Core, SignalR, Angular 21, Kotlin Multiplatform, PostgreSQL

- A multi-provider LLM agent with custom tools does the dispatcher's job: it assigns freight to trucks, checks federal hours-of-service rules, and plans multi-stop routes.
- I built the rest of the product as well: a multi-tenant .NET backend, Angular portals, a Kotlin Multiplatform driver app, and load-board and telematics integrations.

Bookshelf Scanner https://github.com/suxrobgm/bookshelf-scanner
YOLO, Moondream2 VLM, llama.cpp, FastAPI, Angular

- Photograph a shelf and get a book list back. YOLO segments each spine, Moondream2 reads the title and author, and an Angular UI handles corrections and export. Won an Outstanding Project Award at Northeastern.

OGStack https://ogstack.dev | GitHub
Next.js 16, ElysiaJS, Bun, Satori, Cloudflare R2, Stripe

- Uses Claude, GPT, and DeepSeek to read a site's branding and generate its link-preview image. Served 10K+ images from a global CDN during beta.